Linear Transformations

A linear transformation maps vectors by matrix multiplication:

$$\begin{pmatrix} x \\ y \end{pmatrix} \longmapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$$

Linearity means $T(\mathbf{u} + \mathbf{v}) = T\mathbf{u} + T\mathbf{v}$ and $T(k\mathbf{u}) = kT\mathbf{u}$.

For example, $M = \begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix}$. Find the image of $(1, 2)$.

Image

The origin stays fixed because $M\mathbf{0} = \mathbf{0}$.

True or false? 3

Which statements are true?

1. A non-zero translation in the plane can be represented by a 2×2 matrix.
2. A linear map can change the dimension of the space.
3. Every linear transformation fixes the origin.

Decisions

Examples 4

a. Find the matrix of $(x, y) \mapsto (x + 2y, 3x)$.

b. Find the matrix of $(x, y) \mapsto (-2y, 3x + y)$.

a

b

Transform the square $A(1,1), B(3,1), C(3,3), D(1,3)$ using $M = \begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}$.

Vertices

Sketch

1. Which maps are linear?

a. $(x, y) \mapsto (2x, y + 1)$

b. $(x, y) \mapsto (x^2, y)$

c. $(x, y) \mapsto (2x + y, x + xy)$

d. $(x, y) \mapsto (y, -x)$

e. $(x, y) \mapsto (y + 3, x + 3)$

f. $(x, y) \mapsto (2x, 3y - 2x)$

a

4a. Find the matrix of $(x, y) \mapsto (y + 2x, -y)$.

4b. Find the matrix of $(x, y) \mapsto (-y, x + 2y)$.

a

b

7. A rectangle has vertices $A(2,1), B(4,1), C(4,2), D(2,2)$. Apply $M = -I$, sketch its image, and describe the transformation.

7

Sketch

9. Transform the square with vertices $(-1,0), (-3,0), (-3,2), (-1,2)$ using $M = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$. Sketch and describe the image.

9

Sketch

Determining a matrix for a transformation 9

The columns of a transformation matrix are the images of the basis vectors.

$$\begin{pmatrix}1\\0\end{pmatrix} \mapsto \begin{pmatrix}a\\c\end{pmatrix}, \quad \begin{pmatrix}0\\1\end{pmatrix} \mapsto \begin{pmatrix}b\\d\end{pmatrix}$$

Therefore the matrix is $\begin{pmatrix}a & b\\c & d\end{pmatrix}$.

Find the matrix of reflection in the y-axis.

Reflection

Further Examples 10

A rotation through θ anticlockwise about the origin sends the basis vectors to

$$\begin{pmatrix}\cos\theta\\\sin\theta\end{pmatrix}$$

and

$$\begin{pmatrix}-\sin\theta\\\cos\theta\end{pmatrix}$$

Rotation matrix

90 degrees

Interactive transformation exploration 11

Explore how the images of the two basis vectors determine a transformation. Predict the effect before changing the matrix.

Open the interactive GeoGebra activity

Record your observations and test them using a matrix calculation.

Test Your Understanding So Far 12

1. Find the matrix of reflection in $y = x$.

1

2. Find the matrix of rotation 270° anticlockwise about the origin.

2

Further Test Your Understanding 13

Edexcel FP1 January 2009, Q10. Describe the single transformation represented by

$$C = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

Transformation

Invariant points and lines 14

Invariant points

An invariant point maps to itself. Solve $M\mathbf{x} = \mathbf{x}$.

For reflection in $y = x$, every point on $y = x$ is invariant.

For a non-identity rotation, only the origin is invariant.

Invariant lines

An invariant line maps onto itself; its individual points need not stay fixed.

Reflection in $y = x$ preserves $y = x$ and every line $y = -x + c$.

A 180° rotation preserves every line through the origin. Other non-identity rotations have no invariant lines.

Line of invariant points 15

Find the invariant points of $M = \begin{pmatrix} 5 & 4 \\ 2 & 3 \end{pmatrix}$.

Equations

Result

Check by substituting $(x, -x)$ into the transformation.

Finding invariant lines – worked example in class 16

Find the invariant lines of $M = \begin{pmatrix} 3 & 2 \\ 5 & 6 \end{pmatrix}$.

Start with $y = mx + c$.

Images

Invariant lines

6. Let $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

a. Describe both transformations.

b. Find the image of $(3, 2)$ under A.

c. If $B(a, b) = (a - 3b, 2a - 2b)$, find a, b.

a

7. Let $M = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$.

a. Describe M.

b. If $M(p, q) = (-\sqrt{2}, -2\sqrt{2})$, find p, q.

c. Find M^3.

d. Describe M^3.

a

8. Describe $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and find A^{50}.

8

9. A rotation has matrix $\begin{pmatrix} a & b \\ c & -1/2 \end{pmatrix}$.

Find the possible matrices.

9

10. A rotation of $270°$ clockwise about the origin maps (a, b) to $(a - 5b, 4b)$. Find the possible points.

10

Challenge

Derive the general rotation matrix using the images of $(1, 0)$ and $(0, 1)$.

Challenge

Enlargements 20

Describe these transformations:

$A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$

$B = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$

Examples

For positive a, b, $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ stretches by factor a parallel to the x-axis and factor b parallel to the y-axis.

A negative factor also reverses that coordinate; a zero factor collapses a dimension.

Area under a linear transformation 21

The triangle with vertices $(1,1), (1,2), (2,2)$ is transformed by $M = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}$. Find its image and compare the areas.

Areas

Sketch

Area scale factor 22

Area scale factor

Image area $= |\det M| \times$ original area

A negative determinant reverses orientation. Find the image areas below.

a. Area 4, $M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

b. Area 3, $M = \begin{pmatrix} 2 & 0 \\ 9 & 4 \end{pmatrix}$.

a

b

c. Area 9, $M = \begin{pmatrix} 5 & 3 \\ -2 & -1 \end{pmatrix}$.

d. Area 1, $M = \begin{pmatrix} -5 & 2 \\ -4 & -2 \end{pmatrix}$.

c

d

Explore area scale factors 23

Explore how a matrix changes area. Compare the area scale factor with the absolute value of its determinant.

Open the interactive GeoGebra activity

Record your observations and test them using a matrix calculation.

Test Your Understanding 24

Edexcel January 2011, Q8. A shape is transformed by $A = \begin{pmatrix} 2 & -2 \\ -1 & 3 \end{pmatrix}$. Its image has area 72. Find its original area.

Area

5. Describe $M = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$. A triangle of area k has image area 24. Find k.

5

8. The triangle $(a, 1), (4, 1), (4, 3)$ is transformed by $2\sqrt{5}I$. Describe the transformation and find a if its image area is 60.

8

9. Let $M = \begin{pmatrix} p & 1 \\ p & q \end{pmatrix}$, where $q > 0$, and $M^2 = 6I$. Find p, q.

9

11. A pentagon of area 12 is transformed by $M = \begin{pmatrix} -1 & \sqrt{2} \\ -\sqrt{2} & -1 \end{pmatrix}$. Find the image area.

11

12. The triangle with vertices $(k, 1), (4, 1), (4, k)$, where k is an integer, is transformed by $M = \begin{pmatrix} 4 & -1 \\ k & 2 \end{pmatrix}$. Its image area is 10. Find k.

Area equation

Solution

13. Rotate the triangle $(0, 0), (7, 7), (3, -2)$ through $45°$ anticlockwise about the origin. Find its image, determinant and area.

Images

Area

Combining transformations

Apply A first, then B:

$\mathbf{x} \mapsto A\mathbf{x} \mapsto B(A\mathbf{x})$

Order

Matrix multiplication is associative: $B(A\mathbf{x}) = (BA)\mathbf{x}$.

It is generally not commutative: $BA \neq AB$. The rightmost transformation acts first.

1. Reflect in $y = x$, then stretch parallel to the x-axis by factor 4. Find the combined matrix.

1

2. Rotate 90° anticlockwise, then reflect in $y = x$. Describe the combined transformation.

2

Test Your Understanding 31

Edexcel January 2013, Q4. Let P represent rotation 90° anticlockwise about the origin and Q reflection in $y = -x$. Find the matrix and transformation for P followed by Q.

Combined transformation

Exercise 7D Pure Core 1 - P141-143 32

5. Show that reflection in the y-axis followed by reflection in $y = -x$ is a rotation. Give its angle and centre.

5

7. Let $P = \begin{pmatrix} -4 & 0 \\ 0 & 2 \end{pmatrix}$ and $Q = kI$.
Find PQ, describe the transformation, and compare PQ with QP.

7

8. Find and describe A^2, where $A = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}$.
Generalise for $B = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$.

a

b

10. An enlargement of negative scale factor k, followed by a rotation of angle θ, has matrix

$$M = \frac{3}{\sqrt{2}}\begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$$

Find k and $0 \leq \theta < 360°$.

10

11. Let $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $B = 5I$. A shape is transformed by B then A; its image area is 75. Find the original area.

11

13. Show that applying $A = \begin{pmatrix} k & \sqrt{3} \\ \sqrt{3} & -k \end{pmatrix}$ twice is an enlargement.

13

14. Generalise for $P = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$.

14

Transformations in three dimensions

The three columns are the images of $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

Find the matrix of reflection in the plane $z = 0$.

Reflection

The x- and y-coordinates stay fixed; the z-coordinate changes sign.

Rotations in three dimensions

Use the right-hand convention: positive rotation is anticlockwise when looking towards the origin from the positive end of the axis.

$$R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

$$R_y(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$$

$$R_z(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Explore three-dimensional transformations 39

Explore rotations and reflections in three dimensions. Follow the images of the basis vectors.

Open the interactive GeoGebra activity

Record your observations and test them using a matrix calculation.

Test Your Understanding 40

Describe the transformation and find the image of $(-1, -2, 1)$:

$$M = \begin{pmatrix} \sqrt{3}/2 & 0 & 1/2 \\ 0 & 1 & 0 \\ -1/2 & 0 & \sqrt{3}/2 \end{pmatrix}$$

Transformation

Image

3. Let $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$.

a. Describe M.

b. Find the image of $A(3, -1, 4)$.

c. If $(a, -a, 2a - 1)$ maps to $(a, a - 5, -a)$, find a.

a

b

c

4. P rotates 120° anticlockwise about the z-axis (viewed from its positive end).

a. Find P.

b. Find the image of $(3, -1, 0)$.

c. Find the image of $(k, 0, k)$.

a

b

c

5. A reflects in $x = 0$; B reflects in $y = 0$.

a. Find A, B.

b. Find the image P' of $P(a, b, c)$ under A.

c. Transform P' using B.

a

b

c

6. Describe M and find the image of $(k, -k, 0)$:

$$M = \frac{1}{2}\begin{pmatrix} -\sqrt{3} & 0 & -1 \\ 0 & 2 & 0 \\ 1 & 0 & -\sqrt{3} \end{pmatrix}$$

a

b

Inverse matrices for inverse transformations 45

Inverse transformations

If $\mathbf{y} = M\mathbf{x}$ and M is invertible, then $\mathbf{x} = M^{-1}\mathbf{y}$.

$M = \begin{pmatrix} 4 & -1 \\ 3 & 1 \end{pmatrix}$ maps a triangle to $A'(4, 3), B'(4, 10), C'(-4, -3)$. Find its original vertices.

Inverse

Vertices

Test Your Understanding 46

Edexcel June 2012, Q9 (selected parts).

$M = \begin{pmatrix} 3 & 4 \\ 2 & -5 \end{pmatrix}, \quad A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

a. Find $\det M$.

e. Describe A.

f. A followed by B is equivalent to M. Find B.

a

e

f

4. $A = \begin{pmatrix} 1 & 2 \\ -3 & -2 \end{pmatrix}$ maps P to $(5, 8)$.

a. Find P.

b. Applying A then B gives reflection in $y = -x$. Find B.

a

b

5. Let $E = 4I$.

a. Describe E.

b. Find E^{-1}.

c. A triangle has image vertices $(4, 6), (9, 7), (3, 1)$. Find the original vertices.

a

b

c

6. Let $M = \text{diag}(a, b)$, with $a, b \neq 0$.

a. Find M^{-1}.

b. Find the point mapping to $(-6, 8)$.

6

7. Let $R = \frac{1}{2}\begin{pmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{pmatrix}$.

a. Describe R.

b. Find the point mapping to $((1 - 2\sqrt{3})/2, (2 + \sqrt{3})/2)$.

7

8. $A = \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$. A triangle is transformed by AB. Find the matrix that maps the image back.

8

9. $A = \begin{pmatrix} 6 & -2 \\ -4 & 1 \end{pmatrix}$ maps P to (a, b). Find P.

9

10. $M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & -2 \\ 1 & 3 & -1 \end{pmatrix}$ maps (a, b, c) to $(3, 2, -1)$. Find M^{-1} and a, b, c.

10

12. $A = \begin{pmatrix} 1 & 3 & -1 \\ -1 & 1 & 2 \\ 2 & 0 & -1 \end{pmatrix}$. Applying A then B gives reflection in the xy-plane. Find B.

12